

PPG Amyloid PET (FBB) Methods: Processing and Quality Control

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Overview

The following document describes the methods for processing amyloid-PET data using [^{18}F] florbetaben (FBB). These procedures are consistent with the FBB protocol provided by the Jagust Lab at UC Berkeley for ADNI (revised, 01/14/2021). All amyloid-PET scans were acquired on a Siemens Biograph64 TruePoint system (University of Southern California).

Key Points:

- The Desikan-Killiany atlas was used to extract regions of interest (ROIs) for the frontal, lateral parietal, lateral temporal, and anterior-posterior cingulate areas.
- Each ROI and the reference region were registered to PET space using the Advanced Normalization Toolbox (ANTs) in preparation for uptake analysis.
- Standardized uptake value ratios (SUVRs) were calculated by dividing the mean uptake in each ROI by the whole cerebellum. A composite SUVR was calculated to serve as a global measure of amyloid burden. Amyloid status was determined based on thresholds established by the UC Berkeley ADNI protocol, with a composite SUVR of ≥ 1.08 indicating amyloid positivity.

Methods

Software and Tools:

- dcm2niiX (version v1.0.20240202)
- FreeSurfer (version 7.1.1)
- ANTS (Advanced Normalization Tools)
- FSL (FMRIB Software Library)

Processing Steps:

- FreeSurfer Processing
- PET Preprocessing
 - Motion Correction
 - Averaging of Timeframes
- PET-MR Registration
- Calculation of SUVRs:
 - For Predefined Meta-ROIs

Quality Control Steps

Internal QC codes to be used during processing

The following internal quality control (QC) codes are used to evaluate the quality of PET imaging data throughout the processing pipeline. Each scan is assessed based on specific criteria, and the QC code is assigned based on the overall suitability of the scan for further analysis. These codes help streamline the identification of scans that are ready for analysis, require correction, or should be excluded due to quality issues.

RESULT	DESCRIPTION
External Use	
Pass	The scan meets all quality control criteria and is suitable for analysis without modification.
Partial Pass	The data is in an acceptable format. However, minor issues are present. These may include slight under-segmentation of ROIs in ~1/2 slices or minor motion in the PET image.
Fail	The scan is excluded from analysis due to persistent issues after standard troubleshooting - such as severe motion artifacts, Freesurfer parcellation issues due to pathology, or consistent registration failure.
Internal Use	
Requires Fixing	The scan has significant issues, such as heavy under-segmentation of ROIs/reference region, severe motion artifacts, or major PET-MR registration errors. These steps require troubleshooting and reprocessing.

Step 1: FreeSurfer Processing

- Accelerated sagittal MPRAGE (3T) scans are acquired from the LONI-IDA repository and converted from DICOM to NIFTI format using dcm2niix.
- Raw T1-weighted scans are passed into FreeSurfer's recon-all pipeline. The resulting Desikan-Killiany (aparc+aseg) and Destrieux (aparc.a2009s+aseg) atlases are utilized for further processing.

QC Process:

- The atlas is overlaid on the T1 MRI and checked for any major under-/over-segmentations. If the FreeSurfer parcellation covers all areas of cortex and aligns well with the underlying T1 image, the subject is passed (Figure 1). At times, the mask may over-segment into the meninges and under-segment near the temporal lobes. As long as this is not severe, we make a note and pass the subject.
- If significant portions of the mask are missing, this step is repeated. If the segmentation is affected by a pathological condition (e.g., stroke, tumor), the subject is flagged and excluded from further analysis.

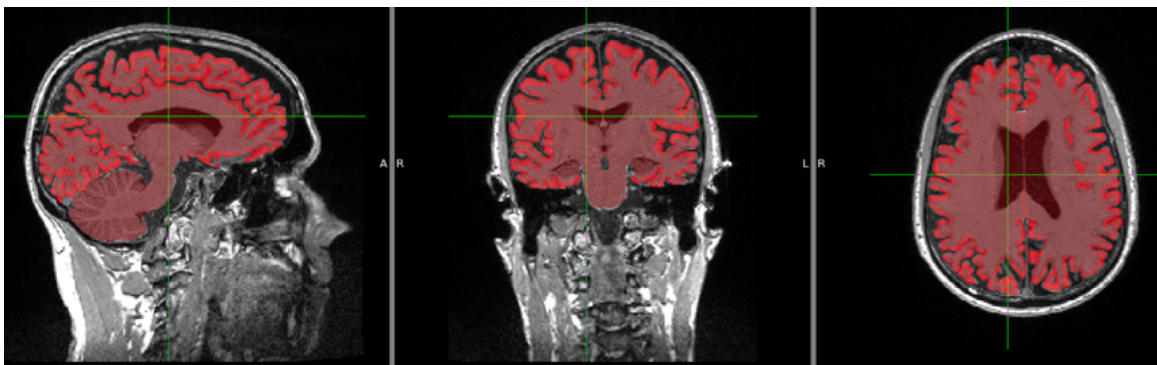


Figure 1: The figure shows the FreeSurfer aparc+aseg mask in T1 space overlaid on the respective T1 MRI.

Step 2: PET Pre-processing

- Amyloid-PET scans 'AA-APOE_AMYLOID (AC)' are acquired from the LONI-IDA repository and converted from DICOM to NIFTI format using dcm2niix. PET scans collected within 8 months of the respective T1-weighted MRI are selected for further processing.
- **Motion Correction:** The PET image is smoothed using a 6mm full-width half-maximum (FWHM) Gaussian kernel. Motion correction is performed using FSL's mcflirt tool, employing normalized mutual information (normmi) as the cost function to optimize alignment between frames. The calculated motion correction transformations are then applied to the original, unsmoothed PET image.
- **Averaging of Timeframes:** Following motion correction, the individual frames of the PET scan are averaged.

QC Process for Motion Correction:

- The motion-corrected PET image is overlaid on the raw PET scan. We scroll through each volume and compare them. The subject passes if no significant distortions or shadows are visible.
- `fsl_motion_outliers` is applied to the motion-corrected PET image to identify outlier frames based on framewise displacement (FD) values exceeding a threshold of 0.9 millimeters.
- If significant distortion and motion outliers exist, the subject is failed.

Step 3: PET-MRI Registration

- The motion-corrected mean PET image is registered to anatomical space using the ANTS registration tool. This process starts with bias correction of the T1 scan via ANTs N4 Bias Correction, followed by skull stripping using FreeSurfer's `mri_watershed` algorithm. The PET image is first linearly registered to the N4-corrected T1, followed by a second round of rigid registration using the skull-stripped brain as the reference image. The transformation matrices obtained from these steps are applied to the motion-corrected mean PET image, which is subsequently smoothed with an 8 mm Gaussian kernel.

QC of PET-MRI registration:

- The smoothed PET image is overlaid on the T1 scan. If the skull, meninges, cortex, and ventricles align well with the native T1 image, the subject is passed (Figure 3). The inverse transformation is also performed (T1 to PET space).

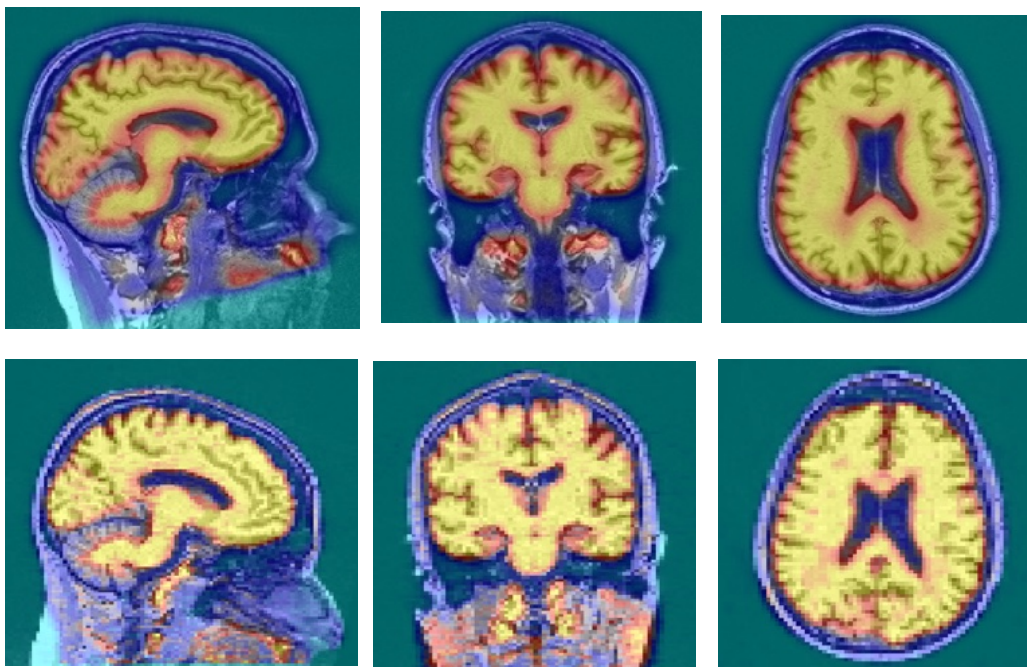


Figure 3: The top panel displays the PET image registered to T1 space (smoothed to aid visualization). The lower panel displays the T1 image registered to the mean PET image.

- Registration issues can result from residual distortions in the PET image due to motion or improper segmentation of the brain mask/skull strip.
- In these instances, the registration can be improved by checking the preprocessing outputs for the PET/T1 data and re-running them.

Step 4: Extraction of Reference Region and Meta-ROI Masks

- For analyses to be conducted in T1 space, the FreeSurfer atlases are first transformed to native space (rawavg.mgz) using `mri_vol2vol`.
- To create the reference region mask, the Destrieux atlas is thresholded to extract voxels corresponding to cerebellar subregions (left and right cerebellar gray and white matter). These are then combined to create a whole cerebellum mask.
- This mask is subsequently binarized and eroded by 1 voxel to reduce boundary inaccuracies caused by partial volume effects (Figure 4).

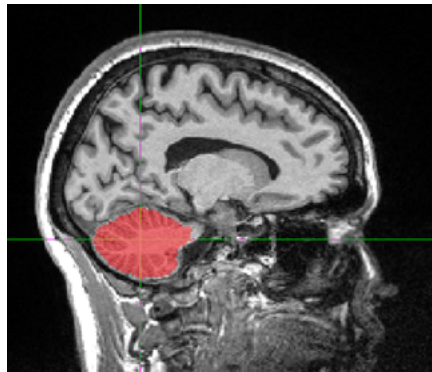
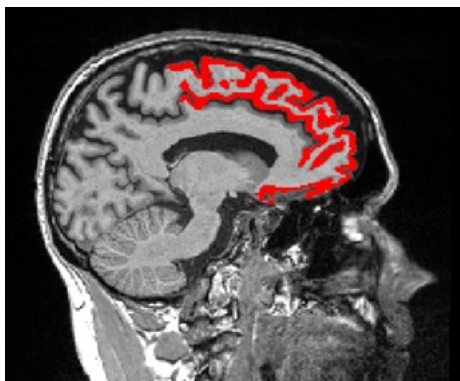


Figure 4: Cerebellum in T1 Space.

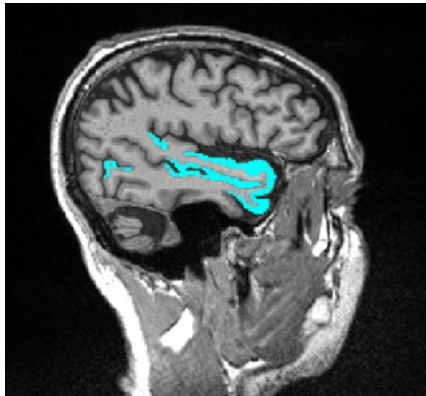
- For the meta-ROI analysis, the Desikan-Killiany atlas is transformed to native T1 space and then passed through a script that extracts voxels belonging to the following regions: Frontal, lateral parietal, lateral temporal, and anterior posterior cingulate.

Frontal ROI

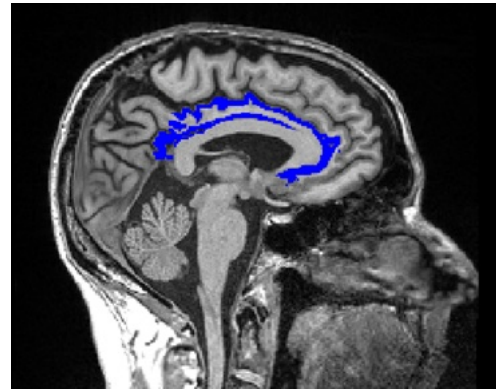


Lateral Parietal ROI





Lateral Temporal ROI



Anterior Posterior Cingulate ROI

Figure 5: Amyloid-PET Regions of Interest (ROIs)

QC for Reference Region and Meta-ROIs:

- The masks are overlaid on the native T1 image and visually inspected for under-/over-segmentations, encroachment into the meninges, or FOV issues that cut any part of the cerebellum off.
- If each mask covers the respective gyri/sulci across all slices, the subject can be passed.
- If a meta-ROI is grossly under-segmented for more than 10 slices, we make a note and rerun this step (Figure 6). Persistent under-segmentations are failed, and processing is halted.

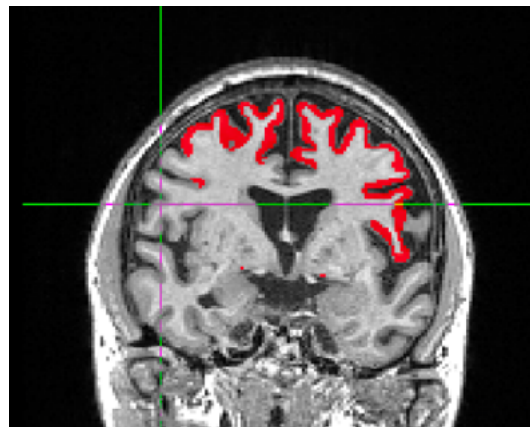


Figure 6: The figure shows the frontal ROI mask overlaid on a T1w image in native space. This ROI is grossly under segmented across multiple slices.

Step 5: Transformation of Reference Region and ROI Masks to PET Space

- The reference region and ROIs masks are transformed back to PET space using the inverse of the PET-to-T1 transformation matrices, employing the Nearest Neighbor interpolation method.

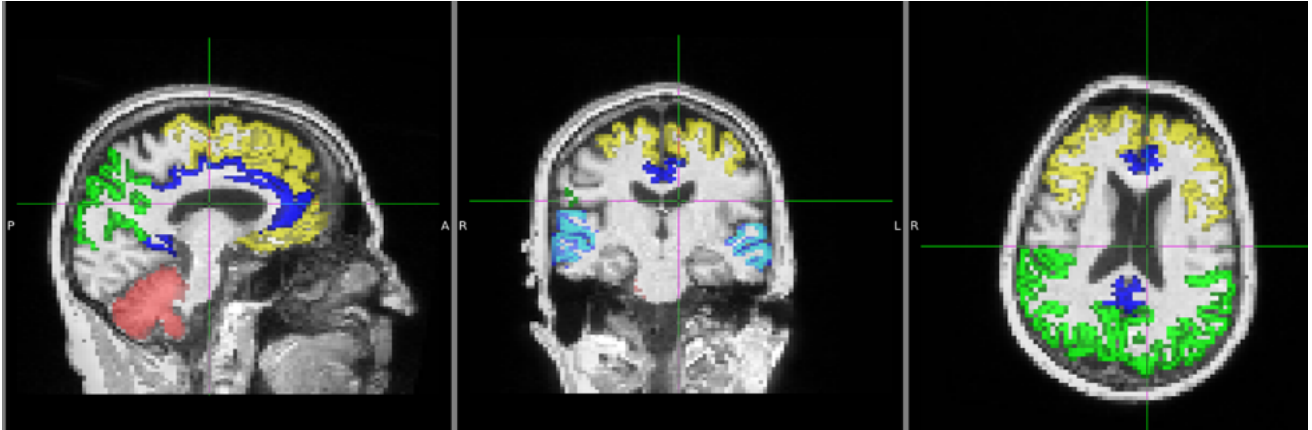


Figure 7: Reference region and meta-ROIs in PET space, overlaid on the T1 image in PET space.

Step 6: Extraction of FBB Standardized Uptake Value Ratios (SUVRs)

- The PET scan is normalized using the whole cerebellum as the reference region.
- FBB Standardized Uptake Value Ratios (SUVRs) for each ROI are calculated by dividing the mean uptake of the ROI by that of the cerebellum.
- The composite ROI SUVR serves as a global measure and is equivalent to the mean of the individual ROI SUVRs.
- The cut-off for amyloid positivity follows the thresholds established by the Berkeley ADNI protocol, designating a composite SUVR above 1.08 as amyloid positive and below 1.08 as amyloid negative.

FreeSurfer Regions (ADNI)

Frontal ROI:

ctx-lh-caudalmiddlefrontal
ctx-lh-lateralorbitofrontal
ctx-lh-medialorbitofrontal
ctx-lh-parsopercularis
ctx-lh-parsorbitalis
ctx-lh-parstriangularis
ctx-lh-rostralmiddlefrontal
ctx-lh-superiorfrontal
ctx-lh-frontalpole
ctx-rh-caudalmiddlefrontal
ctx-rh-lateralorbitofrontal
ctx-rh-medialorbitofrontal
ctx-rh-parsopercularis
ctx-rh-parsorbitalis
ctx-rh-parstriangularis
ctx-rh-rostralmiddlefrontal
ctx-rh-superiorfrontal
ctx-rh-frontalpole

Anterior/posterior cingulate ROI::

ctx-lh-caudalanteriorcingulate
ctx-lh-isthmuscingulate
ctx-lh-posteriorcingulate
ctx-lh-rostralanteriorcingulate
ctx-rh-caudalanteriorcingulate
ctx-rh-isthmuscingulate
ctx-rh-posteriorcingulate
ctx-rh-rostralanteriorcingulate

Lateral parietal ROI:

ctx-lh-inferiorparietal
ctx-lh-precuneus
ctx-lh-superiorparietal
ctx-lh-supramarginal
ctx-rh-inferiorparietal
ctx-rh-precuneus
ctx-rh-superiorparietal
ctx-rh-supramarginal

Lateral temporal ROI:

ctx-lh-middletemporal
ctx-lh-superiortemporal
ctx-rh-middletemporal
ctx-rh-superiortemporal

Dataset Information

Dataset Name	Scans Processed (IDA)	Date Submitted
PPG_Amyloid-PET_FBB	Accelerated Sagittal MPRAGE (3T) AA-APOE_AMYLOID (AC)	3rd December 2024

References

1. Landau, S., Murphy, A., Ward, T., & Jagust, W. Florbetaben (FBB) processing methods.https://adni.bitbucket.io/reference/docs/UCBERKELEYFBB/UCBerkeley_FBB_Methods_01.14.21.pdf
2. Tustison, N.J., Cook, P.A., Holbrook, A.J. et al. The ANTsX ecosystem for quantitative biological and medical imaging. Sci Rep 11, 9068 (2021). <https://doi.org/10.1038/s41598-021-87564-6>